

- 4 Red algae are multicellular photosynthetic protists that contain phycoerythrin. Phycoerythrin is a photosynthetic pigment.

(a) Fig. 4.1 shows:

- the absorption spectrum of phycoerythrin
- the action spectrum of red algae.

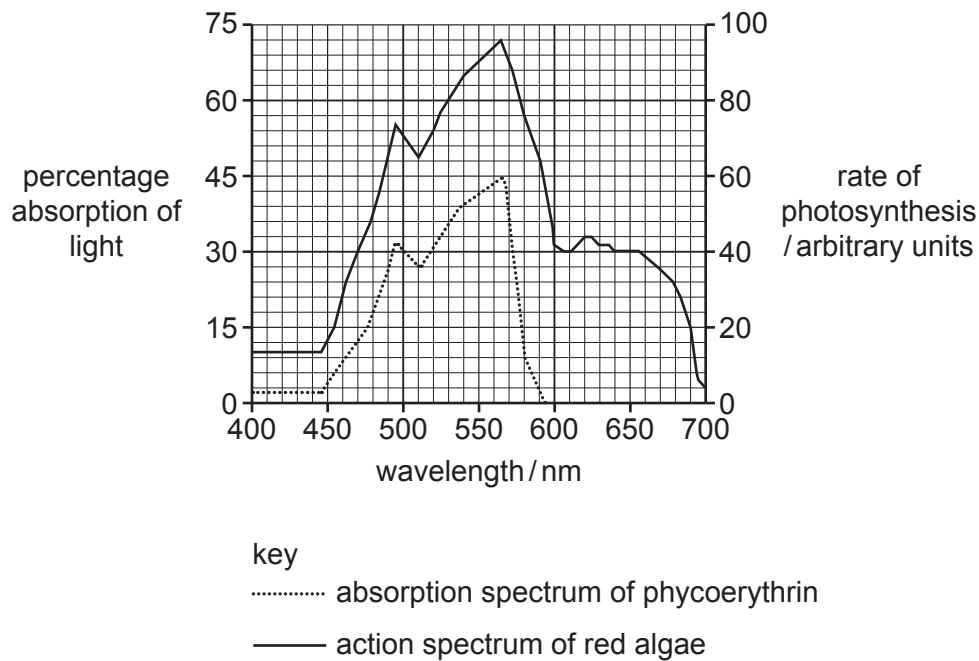


Fig. 4.1

- (i) With reference to Fig. 4.1, state the wavelength of peak absorption by phycoerythrin.

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- (ii) Explain how the data in Fig. 4.1 show that phycoerythrin is **not** the only photosynthetic pigment in red algae.

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..... [2]

- (iii) Phycoerythrin is **not** the primary pigment (reaction centre pigment) for photosynthesis in red algae.

Suggest the role of phycoerythrin in photosynthesis in red algae.

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..... [2]

- (b) Phycoerythrin is rarely found in plants that have green leaves.

- (i) State the name of a technique that can be used to separate and identify photosynthetic pigments.

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- (ii) Explain how the results of this technique would be used to confirm that phycoerythrin is present in red algae and **not** present in a plant with green leaves.

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[Total: 9]